Jingrui Wu

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Education

Rice University — **Houston, TX, USA**
Master of Statistics — **2025.09 – 2026.12 (Expected)**

Hunan University of Finance and Economics — **Changsha, China**
Bachelor of Finance — **2019.09 – 2023.06**

Research Interests

Generative and structure-aware machine learning for single-cell biology and medical imaging; virtual cell models; statistical learning.

Publications & Manuscripts

Peer-Reviewed Publications

1. **J. Wu***, W. Dan*, et al. "Synergy-Aware Residual Flow Matching for Single-Cell Perturbation Prediction." *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2026.
2. W. Dan*, **J. Wu***,†, et al. "Biology-Aware Model Merging for Domain-Adapted Single-Cell Foundation Models." *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2026.
3. S. Hu*, **J. Wu***, H. Ding, F. Zhu, Y. Liu, and S. Xi. "Taming Neural Fields with a Frequency-Lipschitz Curriculum for Sparse-View CT Reconstruction." *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027)*, accepted.

*Co-first author. †Corresponding author.

Research Experience

Semiparametric Latent Variable Modeling with Covariate Adjustment — 2026.01 – Present
Department of Statistics, Rice University; Advisor: Cindy Zhang

- Designed and ran binary and three-class Monte Carlo simulation studies across several levels of class separation, including comparisons with standard clustering methods and the evaluation workflow for class recovery and parameter estimation.
- Developed the real-data analysis used in manuscript preparation.

Dynamic Token Thinning ViTs for Alzheimer's Disease Classification — 2026.03 – Present
Department of Statistics, Rice University; Advisor: Cindy Zhang

- Developing an efficient ViT for CN vs. AD classification on ADNI MRI that adaptively discards uninformative image tokens, combining learnable token scoring with differentiable (Gumbel straight-through) top-k selection, uncertainty-guided early exit, and teacher–student feature distillation.
- Built the end-to-end MRI pipeline for 431 subjects (N4 bias correction, SynthStrip skull stripping, ANTs affine + SyN registration to MNI space) with subject-level train/validation/test splits.

OmniCell Benchmarking and Single-Cell Population Reconstruction — 2026.05 – 2026.08
Abudayyeh-Gootenberg Lab, Harvard Medical School; Advisors: Omar Abudayyeh and Jonathan Gootenberg

- Built a reproducible benchmarking pipeline for OmniCell, the lab's perturbation-aware single-cell foundation model, reconstructing the X-Cell zero-shot setting (380 held-out HepG2 perturbations, 2,000 aligned HVGs from Replogle–Nadig) with checkpoint recovery, gene-space alignment, inference, and matched evaluation.
- Converted mean-level linear-baseline predictions into benchmark-compatible pseudo-single-cell populations (155 K562 perturbations × 8,591 genes) by applying predicted fold changes to real control cells, enabling cell-level evaluation of the baseline.

Sparse-View CT Reconstruction with Frequency–Lipschitz Capacity Control — 2026.01 – 2026.06
Co-first-author research collaboration; publication [3]

- Primary contribution: developed the frequency–Lipschitz capacity-control component, coupling random Fourier feature bandwidth with spectrally normalized MLP capacity to release high-frequency representation power along a coarse-to-fine schedule.
- The joint curriculum improved PSNR by 1.2 / 1.5 dB over the no-curriculum baseline at 60 / 90 views; the full framework achieved the best unsupervised PSNR/SSIM on all four datasets (AAPM, COVID-19, CQ500,

ToothFairy), up to +5.6 dB over Spener, and lost only 0.42 dB under low-dose acquisition (vs. 1.37 dB for SCOPE).

Biology-Aware Model Merging for Single-Cell Foundation Models 2025.12 – 2026.04
Co-first and corresponding-author publication [2]

- Showed that continued-pretraining task vectors in scGPT violate the low-rank assumption of generic merging (25–70% of singular components needed to explain 95% of per-layer energy) and carry weakly supported embedding updates on inactive genes.
- Developed GA-TSV-M, combining spectral-energy-based adaptive rank allocation with gene-activity-aware embedding masking, for data-free merging of four domain-adapted scGPT checkpoints.
- Achieved the best zero-shot AvgBIO among data-free merging methods on all six evaluation datasets (average 0.632 vs. 0.619 for TSV-M), surpassing the individual CPT model on COVID-19, and improved hPancreas annotation Macro-F1 by 12.1 points over TSV-M.

SynGeneFlow: Synergy-Aware Single-Cell Perturbation Prediction 2025.09 – 2026.01
Co-first-author publication [1]

- Developed a conditional flow-matching framework that anchors generation at an additive perturbation prior and learns only the non-additive residual induced by combinatorial genetic or chemical perturbations.
- Designed a perturbation-conditioned gene encoder (static gene–gene mask with dynamic deformation bias) and endpoint- and residual-level MMD objectives for population-level alignment under unpaired single-cell observations.
- On the Norman held-out double-perturbation split, improved DE-Spearman ρ by 11% (0.568 $\rightarrow$ 0.631) and reduced L2 error by 7% over the strongest flow-matching baseline (scDFM); improved synergy Pearson by 18% and non-additive Recall@50 by 19%, with gains transferring to drug combinations on ComboSciPlex.

Technical Skills

Programming: Python (PyTorch, Hugging Face, scikit-learn, NumPy, pandas), R, MATLAB, SQL
Machine learning: flow matching, vision transformers, single-cell foundation models, model merging, distillation
Biomedical data: Perturb-seq single-cell data, MRI preprocessing (ANTs, SynthStrip), CT imaging
Statistics and computing: Monte Carlo simulation, mixture models; Linux, Slurm, Git, LaTeX